Practice Quiz 09D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 09

An image–inverse-image example; a divisibility induction.

1. (a) Define $f : \{1, 2\} \to \{a, b, c\}$ by $f(1) = a$ and $f(2) = b$. Choose $Y \subseteq \{a, b, c\}$ such that $f(f^{-1}(Y))$ is a proper subset of Y, and verify your choice by computing the sets.

   Solution Choose $Y = \{a, c\}$. Only 1 maps into this set, so $f^{-1}(Y) = \{1\}$. Consequently
   $$f(f^{-1}(Y)) = f(\{1\}) = \{a\} \subsetneq \{a, c\} = Y.$$
   The missing element c is not a value of f. ∎

   (b) Prove by induction that $3 \mid (4^n - 1)$ for every integer $n \geq 0$. Here $3 \mid m$ means $m = 3k$ for some integer k.

   Solution For $n = 0$, we have $4^0 - 1 = 0 = 3 \cdot 0$, so the claim holds.

   Suppose $n \geq 0$ and $3 \mid (4^n - 1)$. Choose an integer k such that $4^n - 1 = 3k$. Then
   $$4^{n+1} - 1 = 4(4^n - 1) + 3 = 4(3k) + 3 = 3(4k + 1).$$
   Since $4k + 1$ is an integer, $3 \mid (4^{n+1} - 1)$. This proves the induction step. Therefore $3 \mid (4^n - 1)$ for every integer $n \geq 0$. ∎

Solutions revised using GPT-6 Astra in the Codex harness.